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## AMD (MLDM – Machine Learning and Data Mining) – Module of Practice #02

**Paulo Trigo Silva**

## 1. Extend the Data Model

a) Copy the scripts' folder from the previous practical lesson into a new folder (to use in this lesson).

b) Change the previous scripts with the ones (with the same name) provided in this lesson.

c) Drop all data in your database and execute the new "`02_script_POPULATE_SCHEMA.txt`" file.

d) Confirm that your the database is properly populated (e.g., use "*pgAdmin 4*").

e) Open the "`03_script_CREATE_VIEW.txt`" file and complete the "create view" code in order to obtain the output as presented in the bottom of the file (cf., below the `[OUT]` tag).

## 2. Export Data from the Model

a) Open the "`30_script_EXPORT_DATA.txt`" file.

b) Explore the SQL "`COPY`" statement in order to export the "`v1`" view (cf., previous item) and generate a CSV (icomma-separated values) formatted data file just like "`view_export.txt`". For detailed information on the "`COPY`" statement see "`a01_postgresql-10-A4.pdf`" (page 1394). You may also want to explore the "`\o`" psql command. For detailed information about psql see "`b01_PSQL_withPostgreSQL.pdf`" and "`b02_PSQL_quickReference.pdf`".

c) Explore the psql "`\COPY`" command in order to export the "`v1`" view (cf., previous item) and also obtain the "`view_export.txt`" file (see "`b01_psql_withPostgreSQL.pdf`").

d) Write, in the "`30_script_EXPORT_DATA.txt`", comments on both previous approaches (SQL `COPY` statement and psql `\COPY` command) that describe their main differences.

## 3. Format Data Before Exporting

a) Open the "`03_script_CREATE_VIEW.txt`" file.

b) Build a view, named "`v1_domain`" that projects a single line (tuple) with some constant information; execute and test this new view.

c) Now build a "`SELECT`" statement that combines both the views "`v1_domain`" and "`v1`" so that the "`v1_domain`" line (tuple) always appears on the top of "`v1`" data. You may need to make some (minor) changes on "`v1_domain`" and "`v1`".

d) Use the previous "`SELECT`" statement to build a new view, named "`v1_dataset`". Execute and test this new view.

e) Populate the data model with additional tuples and explore the views' behavior.

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## 4. Dataset (automatic) build of “3RowHeader” Orange Format

a) Consider the "orange data mining" input format described in:

`http://docs.orange.biolab.si/3/visual-programming/loading-your-data/#three-row-header-format`

b) Open the “`03_script_CREATE_VIEW.txt`” file.

c) Implement a view, “`v1_class`” that projects a single line (tuple) that indicates the “class” attribute.

d) Now build a “`SELECT`” statement that combines the views “`v1_domain`”, “`v1_class`” and “`v1`” so that the resulting data matches the "3-Row-Header" input format.

e) Use the previous “`SELECT`” to (re)build the view “`v1_dataset`”. Execute and test this view.

f) Make sure you reproduce data and format presented in “`dataset_3RowHeader.txt`” file.

## 5. Dataset (automatic) build of “HeaderWithType” Orange Format

a) Consider the "orange data mining" input format described in:

`http://docs.orange.biolab.si/3/visual-programming/loading-your-data/#header-with-attribute-type-information`

b) Open the “`03_script_CREATE_VIEW.txt`” file. Follow a similar process as in the previous exercise and build the view “`v2_dataset`” (and any additional views that might be needed) so that so that the resulting data matches the "Header-with-Attribute-Type" input format.

c) Make sure you reproduce data and format presented in “`dataset_HeaderWihType.txt`” file.

## 6. The “Kick-Off” of Final Project A

a) Consider the “final project A” specification (cf., moodle, “Projects” activity).

b) Read project “Goal”, “Scenario A” and “Project Items”.

c) Read “Project Item: 1” and write down (in English or Portuguese) your own “domain description” (do not exceed ½ A4 page).

d) Analyze your “domain description” with your “MedKnow” client.

e) Develop “Project Item: 2”.

f) Analyze your “conceptual data model” with your “MedKnow” client.

g) Develop “Project Item: 3”.

h) Develop “Project Item: 4”.

i) Show your “MedKnow” client a demo of your operational database model and implementation.

j) Develop “Project Item: 5”.